

PAMPLONA AP, Spain

WMO: 080850

Lat: 42.7770N

Lon: 1.6498W

Elev: 458

StdP: 95.94

Time Zone: 1.00 (EUC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.5	-2.0	-7.9	2.0	0.4	-6.0	2.4	0.4	11.5	4.0	10.4	5.0	2.4	340	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.4	34.9	19.7	32.9	19.5	30.7	19.1	21.5	30.1	20.6	29.1	19.9	28.1	3.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	14.5	24.1	18.0	13.7	23.5	17.1	12.9	22.9	64.4	29.8	61.3	29.1	58.7	28.1	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.2	9.1	8.2	-5.8	38.1	1.9	1.5	-7.2	39.2	-8.4	40.1	-9.5	40.9	-10.9	42.0		
			-6.6	23.2	2.0	0.9	-8.0	23.9	-9.2	24.4	-10.3	24.9	-11.7	25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	5.6	6.1	9.3	11.7	14.9	19.5	21.6	21.9	18.6	14.9	9.2	6.1
	DBStd	6.78	3.22	3.34	3.52	3.34	3.60	3.83	3.34	3.44	3.19	3.48	3.35	3.18
	HDD10.0	512	139	113	54	20	4	0	0	0	0	4	53	124
	HDD18.3	2180	393	342	279	199	116	32	9	7	34	115	274	379
	CDD10.0	1730	4	4	33	71	157	284	359	369	259	156	29	4
	CDD18.3	357	0	0	0	1	11	66	110	118	43	8	0	0
	CDH23.3	3682	0	0	1	26	166	753	1080	1172	420	63	0	0
	CDH26.7	1606	0	0	0	2	45	329	508	579	137	7	0	0
Wind	WSAvg	3.3	2.7	3.4	3.5	3.5	3.4	3.6	3.9	3.6	3.1	3.0	3.0	2.6
Precipitation	PrecAvg	1064	109	100	96	108	87	66	44	40	69	99	137	109
	PrecMax	1561	355	270	229	202	200	137	119	114	161	310	239	215
	PrecMin	781	5	20	15	29	31	6	1	6	27	16	35	4
	PrecStd	165	76	64	61	48	39	33	32	30	37	64	54	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.4	17.9	22.2	26.1	30.9	36.2	36.2	37.8	33.7	27.6	19.9	15.2
		MCWB	11.3	9.6	12.5	15.3	17.9	20.2	19.8	20.3	18.8	16.6	14.5	10.9
	2%	DB	13.6	15.3	20.1	23.4	27.2	33.1	34.2	35.0	30.2	25.1	17.6	13.7
		MCWB	10.2	9.9	12.1	14.6	17.1	19.4	19.7	19.7	18.5	16.5	12.9	10.0
	5%	DB	12.1	13.2	18.0	20.9	24.8	30.8	32.1	32.8	28.0	22.9	15.9	12.1
		MCWB	9.2	8.7	11.4	13.3	15.9	18.8	19.6	19.6	17.9	15.7	12.1	9.1
	10%	DB	10.8	11.7	15.9	18.4	22.2	28.1	29.4	30.0	25.8	21.0	14.2	10.9
		MCWB	8.5	8.0	10.3	12.2	15.1	18.3	19.5	19.3	17.2	15.1	11.3	8.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	11.9	14.2	16.6	19.4	21.8	22.5	22.7	20.7	18.6	15.5	11.6
		MCDB	14.3	16.0	20.4	24.2	28.0	30.4	30.9	32.9	29.2	23.9	18.5	14.1
	2%	WB	10.7	10.5	12.9	15.0	17.8	20.6	21.4	21.5	19.5	17.4	13.6	10.4
		MCDB	12.9	14.3	18.5	22.3	25.4	29.7	29.6	30.5	27.3	23.0	16.4	12.9
	5%	WB	9.8	9.4	11.8	13.8	16.6	19.7	20.6	20.5	18.7	16.5	12.6	9.6
		MCDB	11.9	12.4	16.5	19.3	23.2	28.4	28.7	29.1	25.9	21.5	15.2	11.7
	10%	WB	8.7	8.4	10.7	12.6	15.6	18.8	19.8	19.6	17.8	15.7	11.6	8.7
		MCDB	10.4	11.1	15.1	17.5	21.3	26.6	27.6	27.9	24.0	20.0	14.0	10.5
Mean Daily Temperature Range	5% DB	MDBR	7.0	8.5	10.1	10.6	11.6	12.8	13.1	13.4	12.0	10.2	7.4	7.2
		MCDBR	9.2	12.5	14.8	15.9	17.0	17.9	18.3	18.7	16.4	13.8	9.9	8.5
		MCWBR	6.3	7.8	8.0	8.1	7.8	6.9	6.5	6.7	6.6	6.8	6.1	5.7
	5% WB	MCDBR	8.0	11.1	12.5	14.1	15.1	16.3	15.9	16.6	14.3	11.4	8.7	7.6
MCWBR		5.8	7.4	7.2	7.5	7.3	6.8	6.2	6.5	6.1	5.8	5.7	5.5	
Clear-Sky Solar Irradiance	taub	0.303	0.318	0.354	0.377	0.387	0.397	0.387	0.389	0.368	0.345	0.317	0.300	
	taud	2.509	2.458	2.371	2.316	2.320	2.322	2.349	2.350	2.403	2.471	2.508	2.517	
	Ebn at Noon	844	885	887	890	886	875	880	867	861	840	816	813	
	Edh at Noon	71	87	108	123	126	126	121	117	103	86	71	65	
All-Sky Solar Radiation	RadAvg	1.61	2.50	3.72	4.79	5.74	6.53	6.67	5.83	4.57	2.99	1.84	1.46	
	RadStd	0.17	0.29	0.44	0.50	0.42	0.34	0.31	0.30	0.21	0.25	0.19	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page